SyncGuard

Contrastive Audio-Visual Deepfake Detection via Temporal Phoneme-Face Coherence

CS 5330 Computer Vision & Pattern Recognition — Northeastern University, Khoury College of Computer Sciences

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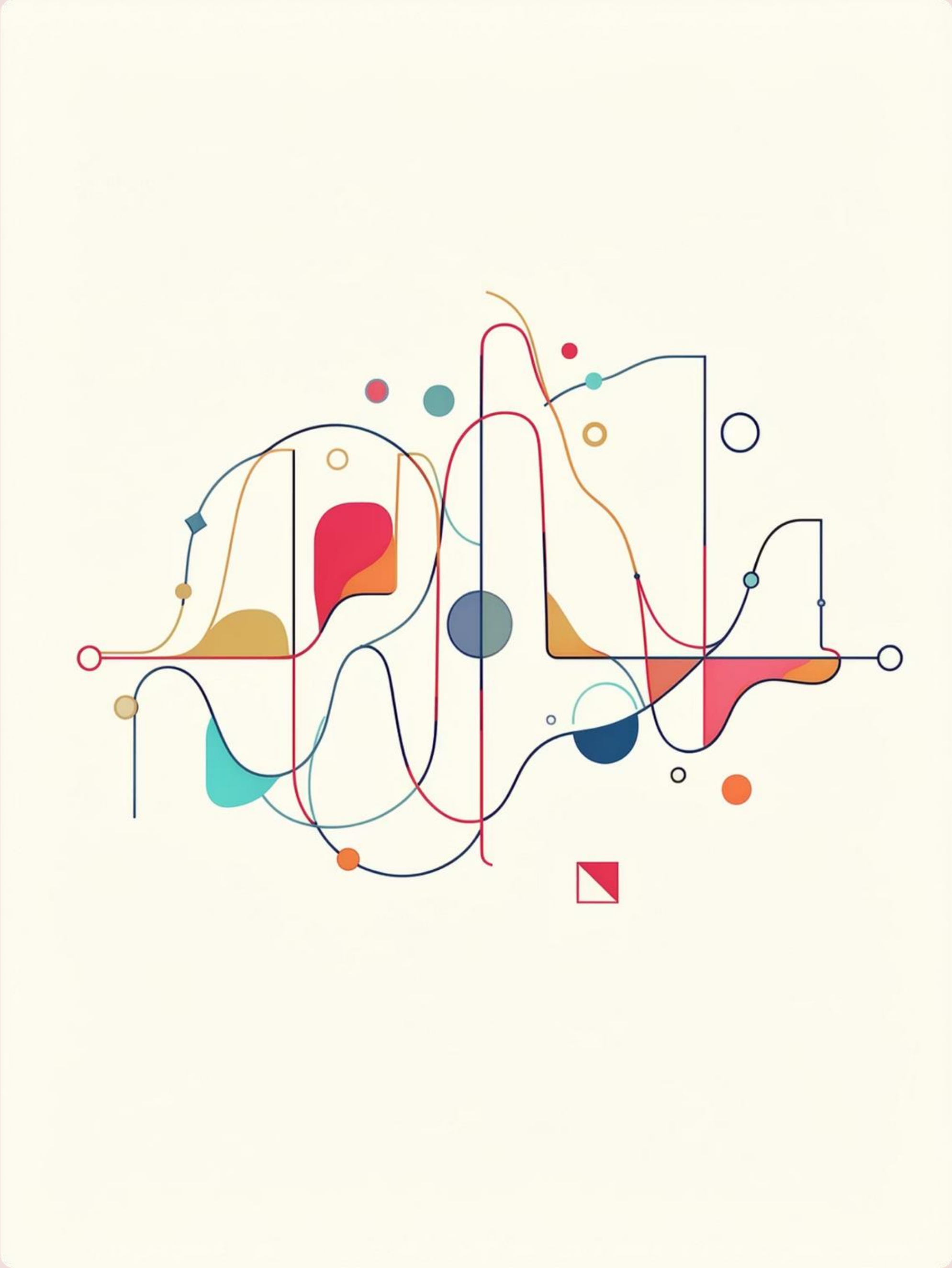


Deepfakes Are Cheap. Detectors Don't Generalize.

Generative AI lowers the barrier

Photorealistic audio-visual deepfakes are now trivial to produce at scale with open-source tools.

Visual detectors fail to transfer

State-of-the-art visual-artifact detectors drop 20–30 AUC points when evaluated across different generators.

Generator-agnostic signals are needed

We need a detection signal that does not overfit to any specific generative architecture or compression artifact.

> "Generator-agnostic detection requires physical, not visual, cues."

Visual artifacts are generator-specific. Physical coupling between lip motion and speech is universal — and cannot be easily forged.

Lip Motion and Speech Are Physically Coupled

Articulatory lead for stops

Lip closure for /p/ and /b/ precedes the acoustic burst by 10–30 ms — a hard biomechanical constraint.

Formant co-occurrence for vowels

Lip rounding for /o/ co-occurs with predictable F1/F2 formant shifts; rounding for /i/ with high F2.

Origin: vocal-tract biomechanics

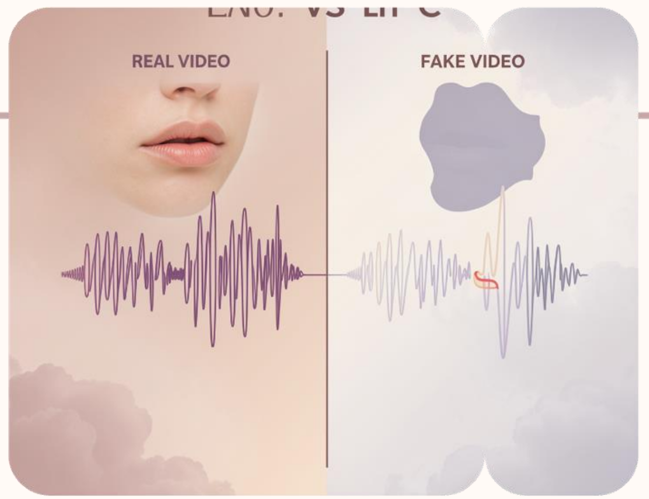
This coupling arises from physical anatomy, not model artifacts — making it robust across generators.

Our core hypothesis

Detectors that learn this coupling via contrastive alignment will transfer across generators without retraining.

Real — Tight Coupling

Lip and audio waveforms aligned in phase

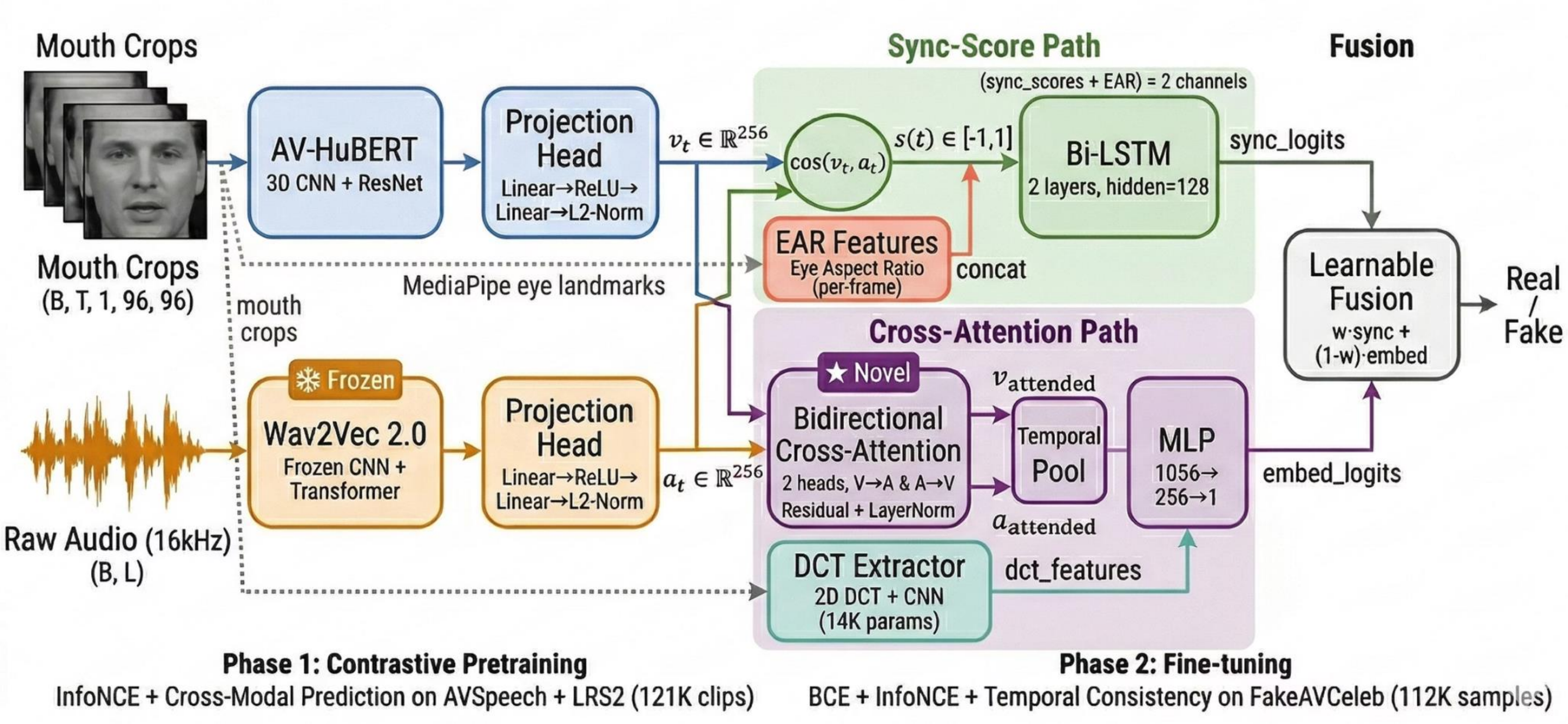


Fake — Temporal Drift

Same waveforms show a visible offset (highlighted)

Real pairs show tight phoneme-viseme alignment; manipulated pairs exhibit measurable temporal drift that crosses the sync-score threshold.

Two-Stream Contrastive Architecture



Visual stream

AV-HuBERT provides lip-reading-pretrained visual features. Each 96×96 mouth crop is encoded to a 256-dim frame-level vector.

Audio stream + fusion

Frozen Wav2Vec 2.0 (layer 9) encodes phonemic content. Per-frame cosine sync-scores feed a Bi-LSTM fused with a learnable cross-attention bypass for voice-clone robustness.

Key Design Decisions

Encoder & Representation

AV-HuBERT over ResNet-18

Lip-reading pretraining yields +4.1 AUC over a ResNet-18 baseline with no pretraining.

Wav2Vec layer 9 selection

Layer 9 corresponds to peak phonemic encoding — earlier layers are acoustic, later layers are semantic.

MoCo queue (4,096 negatives)

Momentum-contrast queue decouples negative count from GPU batch size, stabilizing contrastive loss.

Training & Evaluation Integrity

Frozen Wav2Vec during fine-tuning

Freezing the audio backbone prevents catastrophic forgetting of learned phonemic representations.

Speaker-disjoint splits

No speaker identity overlaps between train, val, and test — prevents face-identity leakage inflating AUC.

Cross-attention bypass

Catches voice-clone attacks (RV-FA) where scalar sync-score alone fails due to matched lip motion.

Preprocessing — Real Video to Model-Ready Features

RetinaFace

Detects faces within the input video or image.

Face Mesh

Generates a detailed 3D mesh of facial landmarks.

Mouth Crop

Isolates the mouth region for focused analysis.

FFmpeg Audio

Processes and extracts the accompanying audio stream.

Visual processing

RetinaFace filters low-confidence frames before MediaPipe extracts 468 landmarks. Eye Aspect Ratio (EAR) flags blinks for temporal masking. Crops are normalized to 96×96 grayscale to reduce color-channel variance.

Audio processing & alignment

FFmpeg resamples all audio to 16 kHz mono; Silero VAD removes non-speech frames. Temporal upsampling bridges the 25 fps video rate to Wav2Vec's 49 Hz feature rate. Throughput: ~190 samples/min with 14 parallel HPC workers.

Two-Phase Training Protocol

Phase 1 — Contrastive Pretraining

Data

AVSpeech + LRS2-BBC — 121,078 real clips, no fake labels required.

Loss

InfoNCE + 0.5 × Cross-Modal Prediction; no BCE at this stage.

Schedule

20 epochs, batch 32, lr 1e-4. Val InfoNCE 8.06; sync-score 0.978.

Phase 2 — Fine-Tuning on FakeAVCeleb

Data

21,544 samples, speaker-disjoint 70/15/15 split across 4 AV manipulation categories.

Loss

InfoNCE + 0.5 × Temporal + 1.0 × BCE (real-only temporal regularization).

Schedule & strategy

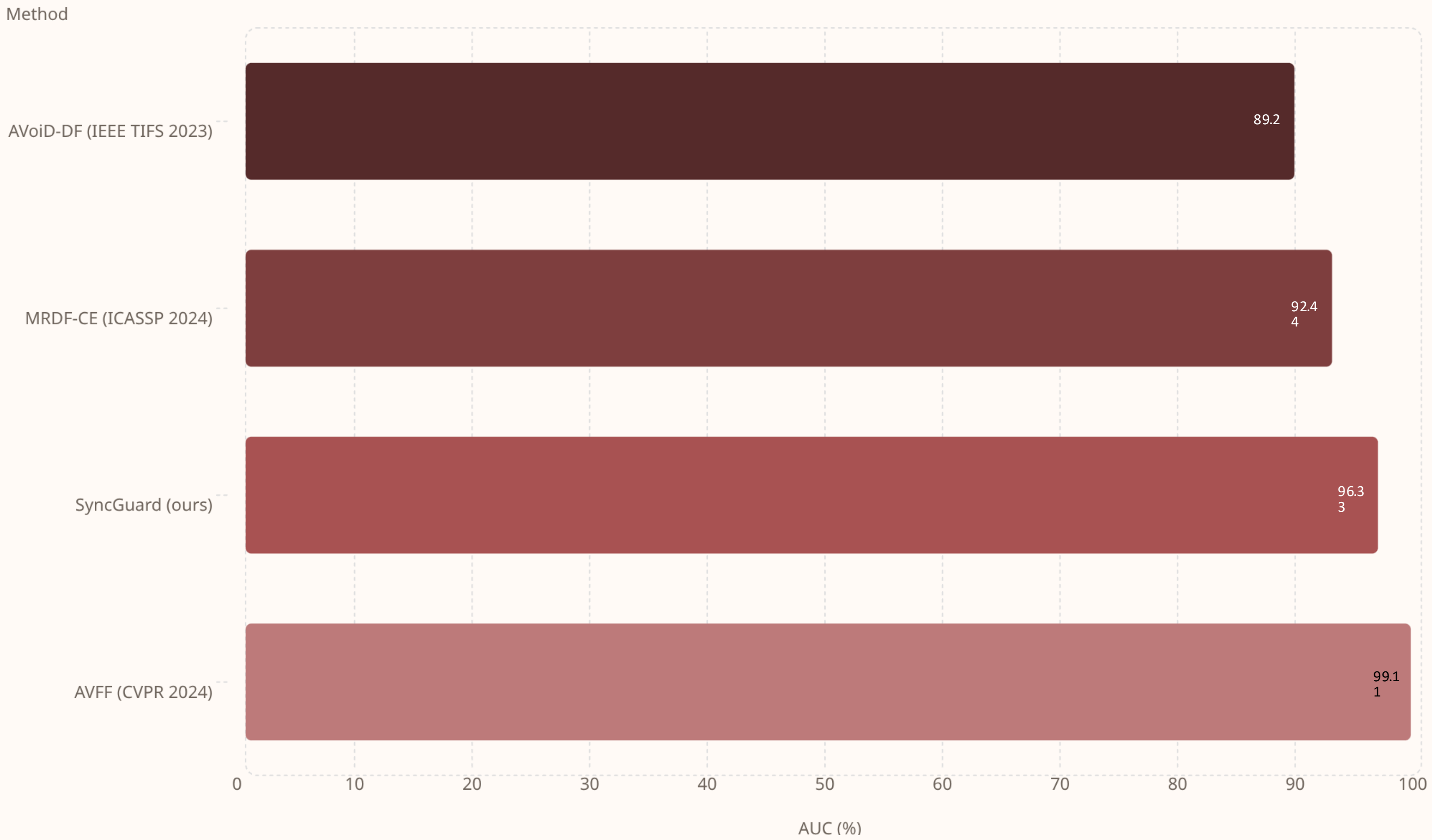
30 epochs, batch 32, lr 5e-5, early stop on val AUC. Cross-attention trained head-only first, then end-to-end unfrozen.

Datasets

Dataset	Samples	Role
FakeAVCeleb	21,544	Primary train / val / test — covers all 4 AV manipulation categories
AVSpeech	24,760	Contrastive pretraining — real clips only, no fake labels
LRS2-BBC	96,318	Expanded pretraining corpus + real anchor pool
DFDC Part 0	1,343	Zero-shot cross-dataset evaluation — unseen generator test
CelebDF-v2	921	Dropped — no paired audio streams; incompatible with AV pipeline

FakeAVCeleb categories: RV-RA (real video, real audio) · FV-RA (face-swap, real audio) · RV-FA (real video, voice-clone) · FV-FA (both modalities manipulated). Speaker-disjoint splits enforced across all partitions.

In-Domain Results on FakeAVCeleb



Our secondary metrics

EER: 9.3%

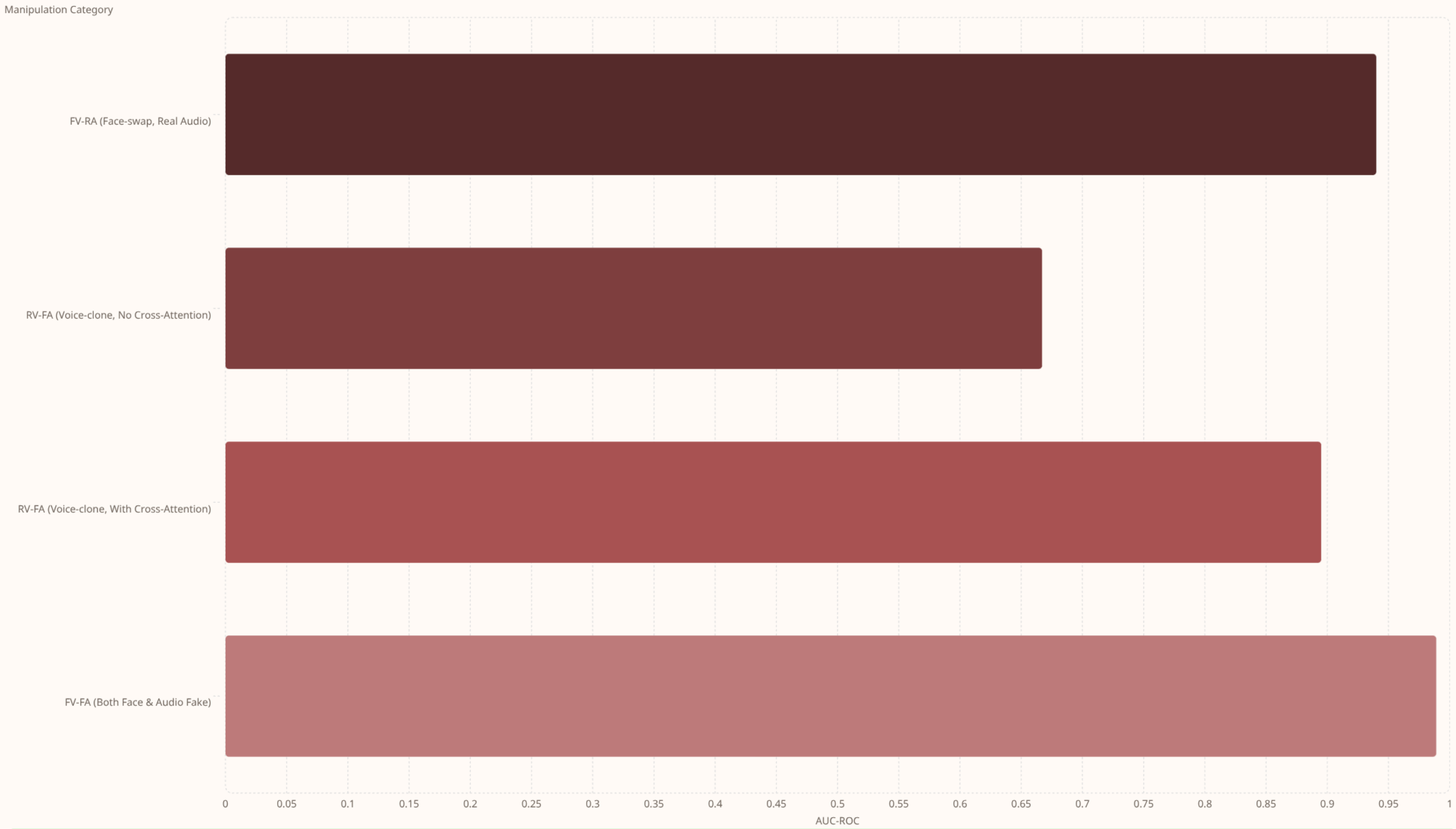
pAUC @ FPR<0.1: 86.1%

FV-FA AUC: 98.9%

*AVFF's 99.1% is likely inflated by leading-silence benchmark bias (Smeu 2025).

FakeAVCeleb test set, n=2,950 samples, speaker-disjoint split. Target was ≥88% AUC.

Cross-Attention Solves Voice-Clone Detection



A **+22.8 percentage point** gain on voice-clone detection (RV-FA) is achieved by integrating the cross-attention bypass, significantly enhancing robustness against audio-only manipulations.

This per-category breakdown highlights the critical role of our cross-attention mechanism in discerning sophisticated voice-clone deepfakes, which traditional sync-score methods often fail to identify. The substantial improvement in RV-FA (Real Video, Fake Audio) scenarios validates our design choice.

DFDC Zero-Shot — A Scientific Negative Result

Our model shows a clear limitation when applied to the DFDC dataset in a zero-shot setting. This reveals a critical challenge for sync-only detection methods.

52.6%

AUC-ROC on DFDC zero-shot evaluation, highlighting a fundamental boundary for sync-score methods against certain attack types.

Underlying Challenges with DFDC

- Face-swap generators preserve original lip motion, maintaining sync between video and audio.
- Sync-score distributions exhibit significant overlap between real and fake DFDC samples.
- Extensive attempts — including BN adaptation, threshold recalibration, and preprocessing fixes — failed to raise AUC past 0.55.
- This indicates a signal problem, not merely a distribution problem; a sync-only framing cannot detect these manipulations.

⚠ This finding underscores that while our approach excels at detecting deepfakes that break audio-visual synchronization, it has inherent limitations against techniques like face-swaps that maintain it. This is not a failure, but a crucial scientific insight into the boundaries of sync-based detection.

What Moved the Needle

Our ablation studies revealed key architectural and training decisions that significantly impacted performance, alongside approaches that proved ineffective or detrimental.

Key Success Factors

AV-HuBERT vs ResNet-18

Lip-reading pretraining in AV-HuBERT boosted AUC by **+4.1** points over a generic ResNet-18 visual encoder.

Cross-Attention on RV-FA

Integrating cross-attention yielded a critical **+22.8** AUC gain for voice-clone detection, addressing a major vulnerability.

Frozen Wav2Vec (Pretrain + Fine-tune)

Keeping the audio backbone frozen was **critical** to prevent catastrophic forgetting and maintain phonemic integrity.

Batch Size 32 vs 16

A larger batch size provided a modest but consistent ~**+0.05** AUC improvement, stabilizing contrastive learning.

Limitations & Ineffective Strategies

DCT Frequency Features

Exploring Discrete Cosine Transform features did **not** yield any performance gain on DFDC, indicating a signal limitation.

BatchNorm Adaptation

Attempts at Batch Normalization adaptation on zero-shot datasets also showed **no improvement** for DFDC, pointing to a deeper architectural mismatch.

Unfreezing Wav2Vec

Allowing Wav2Vec to fine-tune resulted in an **AUC collapse** (from 0.58 to 0.47), confirming the need to preserve its pre-learned representations.

These findings underscore the importance of nuanced architectural choices and robust training protocols for effective deepfake detection, especially in preserving strong audio representations.

Built to Be Reproduced

Our engineering practices prioritize transparency and ease of replication, ensuring that SyncGuard's results can be independently verified and extended by the research community.

Comprehensive Testing

219 `pytest` tests run in 11.8 seconds on CPU, requiring no GPU or dataset, ensuring core functionality remains robust.

HPC Integration

Pre-configured SLURM scripts are provided for seamless deployment on Northeastern's HPC Explorer (H200 GPUs).

Experiment Tracking

Weights & Biases tracking is integrated for every training run, enabling full visibility and auditability of experiments.

Public GitHub Repository

A public GitHub repository offers an end-to-end walkthrough, from setup and smoke testing to full evaluation.

Version-Controlled Parameters

All configurations, random seeds, and preprocessing parameters are checked into version control for complete reproducibility.

Learn more at github.com/Akshay171124/SyncGuard

Five Lessons Worth Carrying Forward

01

Freeze pretrained backbones by default on small downstream datasets. This prevents catastrophic forgetting and leverages robust pre-learned features.

02

Cross-dataset failure is a signal problem first, distribution second. Don't just tweak distributions; investigate if the fundamental signal for detection is present in the first place.

03

Preprocessing parity is load-bearing. One silent 25→30 fps bug invalidated weeks of results, highlighting the critical importance of meticulous data pipeline validation.

04

Auto-resubmit without error gating burns GPU-hours on deterministic bugs. Implement robust error checking to prevent wasteful re-runs of failed experiments.

05

Negative results deserve the same scientific care as positive ones. Document and analyze why certain approaches failed; these insights are crucial for advancing research.

Team & Contributions

Akshay Prajapati

Visual Encoder: Spearheaded the development of the visual encoder architecture.

Preprocessing: Implemented robust data preprocessing pipelines.

Integration Lead: Managed the integration of various model components and the HPC infrastructure.

Ritik Mahyavanshi

Audio Encoder: Designed and optimized the audio encoder.

Contrastive Pretraining: Led the contrastive pretraining efforts to enhance model robustness.

Debugging & Research: Instrumental in NaN debugging and exploring CLIP encoder integration.

Atharva Dhumal

Temporal Classifier: Developed the temporal classification component.

Cross-Attention: Implemented the crucial cross-attention mechanism for audio-visual fusion.

Evaluation Suite: Built the comprehensive evaluation suite and the 219-test Pytest suite for reproducibility.

Thanks to Dr. Akram Bayat • github.com/Akshay171124/SyncGuard